

Uncharted quantum phenomena in Honeycomb Layered Oxides

Layered complex transition metal oxides exhibit exotic crystallochemistry and has numerous applications in the field of high-voltage electrochemistry due to inductive effect, a phenomenon which arises solely due to the layered honeycomb structure of the oxides. High ionic conductivity is required for ultra-fast ionic conductors that could be used as solid electrolytes for use in batterie devices. Existence of alkali atoms in motion within honeycomb slabs of the layered oxides permits them to exhibit high temperature rapid ionic conduction as well as in room temperature and makes them ideal materials to be used in battery and other energy storage systems. These layered oxides offer a playground for the realization of **Kitaev** interaction [1]. The Kitaev model comprises of $S = 1/2$ spins on a honeycomb lattice where the nearest-neighbour interactions are Ising interactions and there are bond-dependent easy axes parallel to the x , y , or z axes. The orthogonal anisotropies of each spin's three nearest-neighbour links oppose thereby resulting in frustration and leading to the formation of an intense entangled spin-frustrated state (spin glass, quantum spin liquids (QSL) etc.). A magnetic field can be used to adjust the fractional excitation of Kitaev QSL, causing them to behave as anions. Braiding these anions is a crucial step in the development of topological quantum computation, indicating that these Kitaev compounds have a lot to offer in the fields of quantum information and technology.

Honeycomb layered oxides have the general chemical formula $A_pM_qD_rO_6$ ($p = 2, 3, 4$; $q = 0, 1, 2, 3$; $r = 1, 2, 3$) where $M = \text{Mn, Zn, Co, Ni, Cu, Cr, Fe}$ and Mg ; $D = \text{Nb, Sb, Ta, Ir, Te, Bi, W, Os, Ru}$ and $A = \text{alkali atoms like Li, K, Na}$ or coinage atoms such as Ag and Cu [1]. Several stacking patterns are adopted by these layered oxides as well as the coordination of the alkali metal atoms and the oxygen atoms around them can be tetrahedral, prismatic, octahedral, and dumbbell shaped. The arrangement of magnetic or even non-magnetic metal atoms in a honeycomb fashion within the slabs of honeycomb layer is the key feature which leads to the emergence of exotic magnetic ground states. The antiparallel spins in some $A_pM_2DO_6$ ($p = 3, 2$) system can take on various configurations like zigzag ordering (observed in **Li₃Ni₂SbO₆**, **Na₂Co₂TeO₆**) or alternating stripe-like patterns inside the honeycomb slab. Spin-flop magnetism is observed in **Na₃Co₂SbO₆** and **Li₃Co₂SbO₆**. Phase diagram of **Na₃Co₂SbO₆** shows that the spin-liquid state is encircled by ferromagnetic states that have moments along c -axis as well as ab plane possessing zigzag type phases having moments in ab plane (ZZ1), along Co-O bonds (ZZ2) and in ac -plane (ZZ3) [2]. **Li₄FeSbO₆** is more fascinating compound, which have an antiferromagnetic ground state. But a magnetic field induces an intermediate magnetic state (II). As a result, there are at minimum two phases associated with two distinct spin orientations. At high pressure, **Na₂Cu₂TeO₆** produces new bonding interaction (Cu bond dimerization) which leads to a shift in magnetic characteristics. **Li₃Mg₂OsO₆** shows spin freezing below 12 K while its sister compound **Li₄MgOsO₆** shows a dominant glassy behaviour below 2 K. Topology of the Ru^{3+} $S = 3/2$ sublattice in **Li₃Mg₂RuO₆** is unusual, which consists of chains of edge-sharing triangles joined by corner sharing, indicating the possibility of geometric frustration. The physical properties of **Li₃Mg₂RuO₆** demonstrate that a long-range AFM ordering near 17 K occurs in the system and provide signature for short-range spin correlations above T_N . Kitaev's nearest-neighbour-only interaction model can be realised under better conditions as one move from $4d$ Ru to $3d$ Co because magnetic d orbital gets more localised. In short, the honeycomb structure of these layered oxides provides the ideal geometry prerequisite for frustration thereby preparing the playground for the advent of enigmatic states and hence the emergence of innovative quantum materials which may prove to be indispensable for future generation quantum computers and may go a long way in upliftment of the human society.

Reference:

- [1] G.M. Kanyolo et al., Chemical Society Reviews 50, 3990-4030 (2021).
- [2] H. Liu et al., Physical Review Letters 125, 047201 (2020).